

Project (If Applicable)		TRA Ref. No.	DOUS-TRA-007	
Location / Worksite				
Sub Location / Area				
Equipment Spread (Optional)				
Task / Activity	UNSTACKING & STACKING SYSTEM			
Approver				
Responsible Person				
Date TRA Created				
Reference Supporting Documents & Task Plan				
S3RA:			Hot Work Checklist:	
Task Plan:			Work at Height Checklist:	
Lift Plan:			Confined Space Work Checklist:	
PTW:			Electrical Equipment Checklist:	
COSHH / SDS:			Electrical Isolation Checklist:	
Rescue Plan:			Machinery Isolation Checklist:	
Work Equipment Assessment:			Pressure Systems Checklist:	
Other:			Other:	
Reason for TRA (Tick where Applicable)				
Identification of New Task	Annual Review	Review following Incident	Management of Change Record	Other, Please Specify
☐	☐	☐	☐	☐
Other (Details):				
TRA Participants				
Name	Position		Signature	

No.	Task /Activity	Hazard and Consequence		Initial Risk			Mitigating Controls	Residual Risk		
	Sequential Steps	Hazard	Consequence	Probability	Consequence Severity	Risk Rating	Measures to Reduce Risk to ALARP	Probability	Consequence Severity	Risk Rating
1	Review TRA and TBT	Miscommunication	Personnel Injury / Environmental Pollution	1	2	M	Situational awareness and proper use of PPE. Hard hats, impact gloves, cotton gloves, latex gloves, steel toed boots, safety glasses, side shields and face shield	1	1	L
2	Prior to any task being undertaken ensure that all personnel are aware of the hazards involved.	Electrical shock, possible slips trips and falls. Pinch points poor body position. Hydraulic discharge and spills.	Personnel Injury / Environmental Pollution	2	3	M	All personnel must be aware of their surrounds and weather conditions. Use proper Manual Handling techniques, i.e., Always use proper body positioning, lift with legs, and obtain second personnel when loads are over 40lbs. Read all SDS on chemicals used, use PPE that is specified in SDS and be aware of where all spill kits are located.	2	2	L
3	Check weather condition prior ROV leaving the deck	Damage to equipment from weather patterns	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Communication with bridge personnel prior to lifting system from deck.	1	2	L
Non-Personnel are not to be allowed around ROV during this operation										
4	Energize HPU on winch and ensure that cooling system on and is operational.	Damage to Winch HPU from overheating	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Either air cooled or water cooled ensure that it is on and functioning correctly.	1	2	L
Perform Steps 5 thru 11 if ROV is on Stands										
5	Ensure that grounding strap has been removed from ROV.	Electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Ensure that ROV is not powered up without grounding strap on.	1	2	L
6	Erect LARS to bring ROV off stands to allow room for TMS and ROV to fit on LARS.	Equipment Damage, burst hydraulic hose on winch, leak of hoses.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Operator and observer must communicate to ensure that ROV is aligned properly and will not damage equipment. Have spill kit available.	1	2	L

7	Bring ROV off deck and remove stands	Pinch points while removing stands from ROV.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Observer will remove the stands one at a time ensuring that they do not place themselves in a pinch point.	1	2	L
8	Slowly bring the ROV/TMS into the docking head and open the latches and latch ROV/TMS into docking head.	Burst Hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Spill kit available and ready for use. Keep HPU in Low Tension while hauling in.	1	2	L
9	Position ROV/TMS in proper location to lower and maintain room for TMS.	Burst Hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Spill kits available and ready for use.	1	2	L
10	Unlatch ROV/TMS from docking head and slowly lower to LARS deck.	Burst Hydraulic hose, leak of hoses, hydraulic discharge. Pinch points and damage to equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Personnel will keep in constant communications as ROV/TMS is being lowered. Keep HPU in Low Tension while hauling in.	1	2	L
11	Observer will attach grounding strap to ROV and secure ROV to LARS system.	Electrical shock, damage to equipment	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Ensure ROV is not powered up without grounding strap. Ensure that ROV is strapped down to LARS system with large approved, cargo straps.	1	2	L
12	Pilot will turn on TMS IPDS.	Electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Observer will ensure Pilot that on one is around or on TMS before activating system.	1	2	L
13	Winch operator will bring up TMS as Pilot unlatches TMS from ROV.	Damage to tether and equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator will be in constant communication with Pilot to unlatch TMS, and not 'Stretch' the tether. Pilot will only pay out only what is needed to complete task. Keep HPU in Low Tension while hauling in.	1	2	L
14	Place TMS into docking head latches	Burst hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Spill kits available and ready for use.	1	2	L

15	Position TMS in proper location by bringing LARS system into proper position.	Burst Hydraulic hose, leak of hoses, hydraulic discharge.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Spill kits available and ready for use.	1	2	L
16	Unlatch TMS from docking head and slowly lower to stands on LARS deck.	Burst Hydraulic hose, leak of hoses, hydraulic discharge. Pinch points and damage to equipment, and tether	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Winch operator and observer will monitor LARS system for leaks. Personnel will keep in constant communications as TMS is being lowered. Keep HPU in Low Tension while hauling in.	1	2	L
17	Land TMS on stands.	Swinging TMS impacting LARS or personnel, Tether damage and equipment damage.	Personnel Injury / Environmental Pollution / Damage to Equipment	3	4	M	Observer must be in constant communication with winch operator. Observer must be aware of all pinch points.	1	2	L
18	Pilot will disable or turn off TMS depending on task to be done.	Electrical shock	Personnel Injury / Environmental Pollution / Damage to Equipment	3	3	M	Pilot will disable or turn off TMS to prevent injuries to personnel and equipment.	1	2	L
19	Observer will attach grounding strap to TMS and secure TMS to LARS system	Electrical shock, Damage to equipment.	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Ensure TMS is strapped down to LARS system with large approved, cargo straps.	1	1	L
20	De-energize HPU on winch and ensure that cooling system is off.	Damage to Winch HPU from overheating	Personnel Injury / Environmental Pollution / Damage to Equipment	2	3	M	Ensure that the Winch has been turned off and secured.	1	1	L

TRA Approval	Name	Position	Signature	Date
By signing, the Approver confirms that the TRA has been reviewed and considers the task acceptable to proceed with residual risk meeting the acceptance criteria of DO Risk Matrix.				

*Reference DO-HSE-FRM-140 for DeepOcean Risk Assessment Matrix (RAM) and Guidewords